

GPU Programming with OpenMP Exercise

Logging on, Compiling and Running

1 Introduction

The purpose of this exercise is to check you can compile and run simple GPU programs on ARCHER2.

2 Connecting to ARCHER2

Log on to ARCHER2:

```
ssh username@login.archer2.ac.uk
```

Note: for this exercise, it is important to use the `/work` file system and not the `/home` file system:

```
cd /work/project/project/username/
```

3 Compiling and Running

The code is contained in `saxyp.tar`. The tar file can be fetched from GitHub by cloning the course repository with the following commands:

```
git clone https://github.com/EPCCed/archer2-GPU-directives.git
cd archer2-GPU-directives
git checkout 2026-10-06
cd Exercises
```

Alternatively, the file can be found on ARCHER2 and copied into your `/work` directory with the command:

```
cp /work/z19/shared/GPUdir/saxpy.tar .
```

Now unpack the file:

```
tar -xvf saxpy.tar
```

4 Compiling and Running

There are two examples, one using OpenMP and the other HIP. To compile and run the HIP example:

```
cd saxpy
sbatch ompsaxpy.slurm
```

and similarly for the HIP example (with `omp` replaced by `hip`).

Note that compilation takes place within the Slurm script as the compilers are only available on the GPU nodes.